

Data Storage & Cloud computing.

1] Intro to Enterprise Data storage

- It is a centralized repository designed for storing business-critical data.
- It provides data sharing, data management, and protection across multiple computer systems.
- It ensures high availability, redundancy & scalability for handling large volumes
- Types
 - Storage Area N/W (SAN)
 - N/W attached storage (NAS)
 - Direct attached storage (DAS)

① Direct Attached Storage :-

- It refers to HDD or SSD's directly connected to a single computer or server.
- This connection can be internal or external.
- DAS is not networked via Ethernet or Fibre channel, means it cannot be accessed by other system.
- The storage device can include multiple drives configured within a server.
- eg :- hard drives, optical disc drives, etc.

Advantages:-

- less expensive compared to NAS
- Ideal for localized file sharing.
- Suitable for storing critical data locally.
- Cheaper compared to NAS or SAN

Disadvantages:-

- Expanding storage in DAS is difficult.
- It is tied to specific server, lacking n/w access for other system.
- Unused storage cannot be reallocated easily.
- If server fails, access to data become impossible



Study material provided by: Vishwajeet Londhe

Join Community by clicking below links



Telegram Channel



https://t.me/SPPU_TE_BE_COMP

(for all engineering Resources)



SPPU Engineering & Technical
UPDATES
WhatsApp channel



WhatsApp Channel

(for all tech updates)



[https://whatsapp.com/channel/
0029ValjFriICVfpcV9HFc3b](https://whatsapp.com/channel/0029ValjFriICVfpcV9HFc3b)



Insta Page

(for all engg & tech updates)

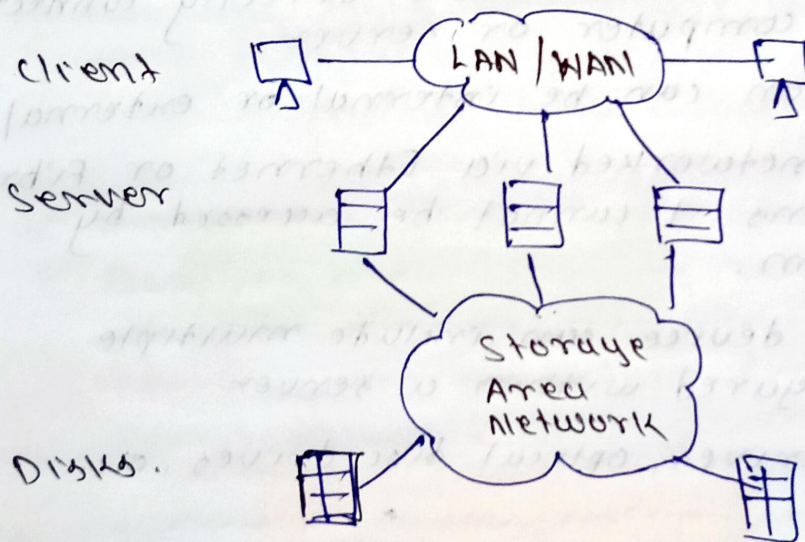


@SPPU_ENGINEERING_UPDATE

[https://www.instagram.com/
sppu_engineering_update](https://www.instagram.com/sppu_engineering_update)

② Storage Area Network (SAN)

- SAN is a high performance, dedicated n/w that connects storage devices to servers.
- It interconnects solid-state storage & shares them with multiple servers.
- SAN is typically used to provide consolidated, block-level data storage.
- It primarily used to enhance storage devices like disk arrays, tape libraries & RAID n/w.
- Commonly used in environments where multiple servers need access to the same data.



Features

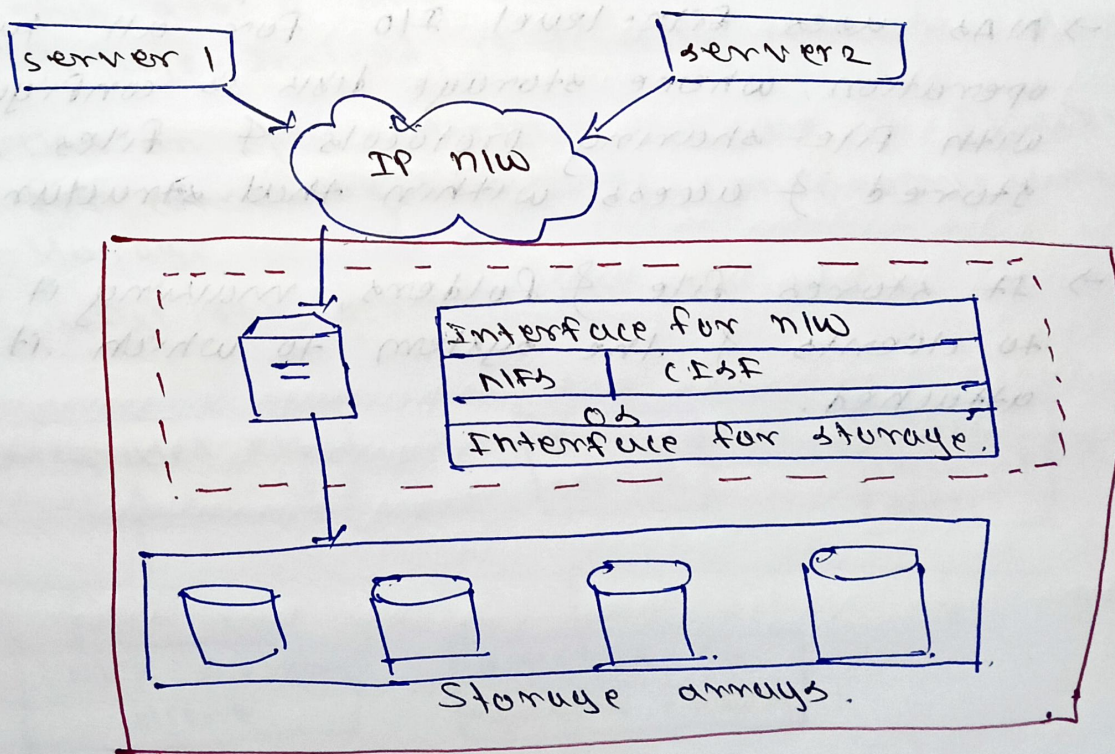
- n/w resources are dedicated to storage only.
- Uses SCSI protocol for end-to-end communication.
- provides block-level access to data rather than file-level, ensuring better performance.

Advantages:-

- Supports large block data transfer.
- provides reliable data transfer
- offers configuration flexibility.
- High performance with low latency.
- Centralized storage management.

③ Network-Attached Storage:-

- NAS is file-level computer data storage server connected to a computer nw that provides data access to a heterogeneous group of clients.
- It is specialized for serving files by either it's h/w or s/w.
- NAS device are storage arrays or gateways that support file-based storage protocols like NFS & CIFS



Components of NAS

① Network interface card (NIC)

② Operating system

③ CPU & memory

④ NFS & CIFS

⑤ Storage protocol

→ NFS :- Network file system & common internet file system protocols used for file sharing.

→ Network interface card is used for providing n/w connectivity.

→ NAS devices use industry-standard storage protocols such as ATA, SCSI, & fibre channel for connecting & managing physical disk storage.

NAS File I/O :-

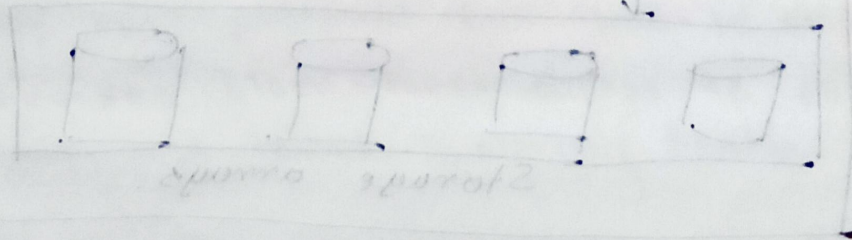
→ In NAS there are two file-level protocols.

(i) CIFS :- allow windows users to share, store & retrieve files from NAS device

(ii) NFS :- allow linux

→ NAS uses file-level I/O for all types of operation, where storage disk is configured with file-sharing protocols & files are stored & access within that structure

→ It stores file & folders, making it accessible to clients & the system, to which it is attached.

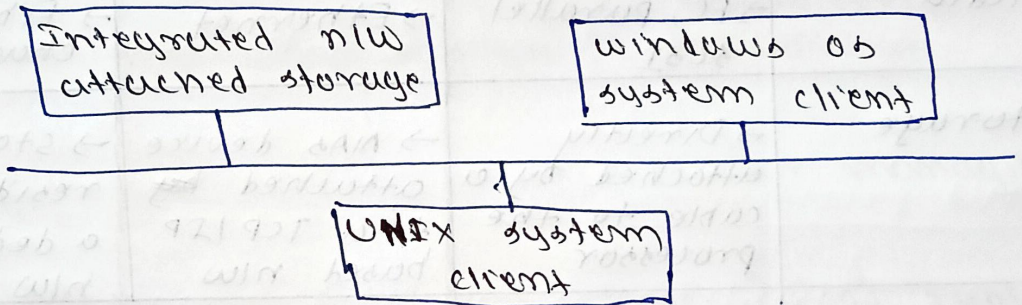


NAS Implementation.

→ NAS can be implemented in two ways:

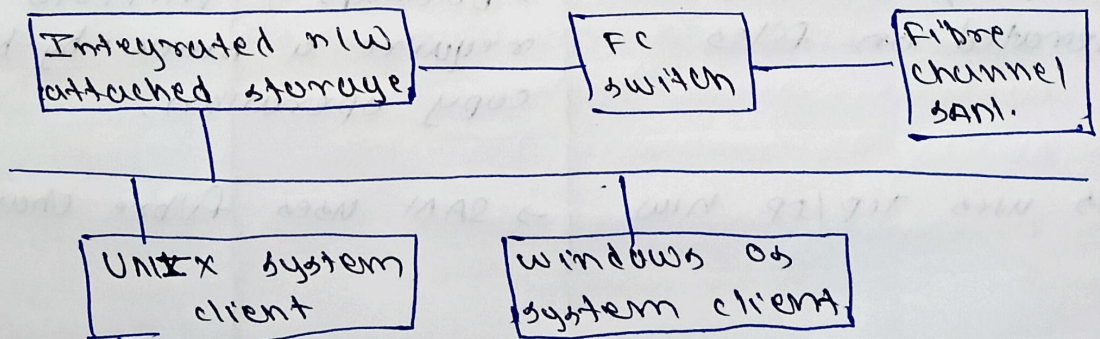
① Integrated NAS :-

- contains all components of NAS in a self-contained environment.
- Suitable when looking for an integrated NAS with its own storage.



② Gateway NAS :-

- This is a NAS device without integrated storage. Instead, it directly connects storage.
- Uses an open interface, making it flexible for connecting to shared, non-proprietary storage.
- Useful for scalable storage needs.



parameter	DAS	NAS	SAN
→ I/O protocol	→ SCSI	→ NFS & CIFS	→ SCSI
→ Data sharing	→ Not supported	→ Supported	→ Supported with specialized SW.
→ Capacity sharing	→ manual	→ Yes	→ Yes
→ media	→ FC, parallel SCSI	→ Ethernet	→ Fibre channel.
→ Storage	→ Directly attached by a cable to the processor	→ NAS device attached by to a TCP/IP based NW	→ Storage resides on a dedicated NW.

NAS	SAN
→ machine connected with LAN may use NFS, CIFS or HTTP to connect to NAS	→ server-class device equipped with SCSI & fibre channel adapters connect to a SAN.
→ The file system is managed by NAS head unit	→ The SAN server manage the file system.
→ Backups & mirrors are generated on files	→ Backups & mirrors require a block-by-block copy operation.
→ NAS uses TCP/IP NW	→ SAN uses fibre channel.

2] Data storage management:-

- A large no. of databases created in private and government organizations & the volume is growing out of control.
- These databases can be difficult to maintain.

Tools :-

- Storage monitoring slw helps users track storage capacity & assign it to various application
- Regular monitoring is crucial for managing storage devices.
- Storage Resource management (SRM) tools help manage storage capacity, monitoring, cloud storage, SAN, NAS, & backup tasks.
- These tools include!
 - ① configuration tools : Set up storage resources & organize RAID device
 - ② provisioning tools : Define & control access to storage resources.
 - ③ measurement tools : Analyze ~~data~~ ^{performance} based on behavioral ~~info~~ ^{info} about storage device.

Process :-

- ① start : Data is stored based on importance and access freq.
- ② backup & recovery : Regular backups are made with recovery plans to restore data
- ③ Resource management : Managing resources for backup data, such as storage media.

cloud provisioning :-

- The allocation of a cloud provider's resources, and services to customer
- Key cloud services include IaaS, SaaS, PaaS in both public & private environments.
- process involves preparing, reserving, accessing, using & releasing resources.

Benefits :-

- ① continuous improvement through process measurement
- ② Isolation of installation, configuration & customization tasks.
- ③ Ability to measure progress
- ④ Assembly line approach.

characteristics:-

- missing / incorrect info is resolved.
- The product is delivered to the customer before final provisioning.

Types:-

① Dynamic provisioning:-

- Resources are scaled up/down based on demand, billing based on usage.

② Self-provisioning:-

- Customers buy resources via a web portal using self-service tools.

③ Advance provisioning:-

- Customer contact in advance for required services.

3] File system:-

- File system is the interface b/w secondary storage & user's application.
- Its purpose is to maintain a consistent view of storage, allowing users to manage files.
- It also ensures file security through access control lists. (ACL's)

① FAT File System

- The File Allocation Table File system is simple, designed for small disks & basic folder structure.
- It operates using file system blocks, where modern systems use block sizes of 512 to 65,536 bytes.
- Windows use FAT & NTFS file system.
- FAT12, FAT16 & FAT32 denote the number of bits used per file system block.
- Linux uses Ext2, Ext3, Ext4 file system.

② NTFs :-

- NT file system is used by windows NT for efficient storing, organizing & finding files on hard disk.
- NTFs supports larger file sizes, larger hard drive and is more secure than FAT.
- The cluster size in NTFs ranges from 4 to 128 kilobytes.
- NTFs allows file & folder compression to save disk space for more efficient usage.

③ Cloud File system :-

- cloud storage is an abstraction of storage resources over the internet.
- It allows for pooling & sharing of resources.
- File storage in cloud is hierarchical & is structured to organize data.
- This system is most suitable for unstructured or semi-structured data.
- cloud file storage works on pay-per-use basis & follows common protocols like CIFS, NFS.
- It is scalable, and ideal for organizations to manage different workloads.

Benefits :-

- Scalable
- Interoperability.
- Budget & resources.

5] Using Grids for data storage:-

- Grid computing refers to a distributed computing system where multiple computers are connected to form a large virtual system.
- Grid Storage is a type of data storage that uses n/w of interconnected nodes to create scalable & versatile storage soln.
- Grid computing involves physically distributed computer resources, often managed by diff. organization, working co-operatively to run specific application.

Grid oriented Storage (GOS):-

- It is a specialized data storage architecture designed to connect directly to a computer computational grid.
- It facilitates advance data banking services, allowing data to be shared & accessed by multiple computers.
- Gos is an evolution of NAS, tailored for Grid computing.
- It optimized for handling long-distance, cross domain & single-image operations.

Features

- Distributed storage for managing structured data
- Uses Google File system (GFS) for storing log & data files.
- Uses chubby for distributed lock services.
- Supports single-row transactions.

② Dynamo ↓

- Dynamo is Amazon's key-value structured storage system.
- It can act as a database or a distributed hash table
- primarily designed for highly available, scalable applications.
- Dynamo uses consistent hashing to distribute data.
- Dynamo provides eventual consistency & relies on a decentralized system.

4] Cloud data Stores :-

- A data store is a connection to a store of data, regardless of whether data is stored in database or in files.
- A data store acts as a repository for storing, managing & distributing datasets at an enterprise level.
- Data store data can be structured, unstructured or in another electronic format.

Distributed data store:-

- A distributed data store refers to a computer n/w where info is stored across multiple nodes.
- A distributed data system stores files, scripts and image across more than one server.
- eg → Google's BigTable, Amazon's Dynamo.
- These data stores are typically non-relational databases designed for fast data searches.

Types

(i) BigTable :-

- It is a distributed storage system designed for managing & storing structured data at Google.
- It can reliably scale to petabytes of data across thousands of machines.
- It stores data as rows, columns & timestamps in a dynamically partitioned system called "tablets".

cloud file systems +

① Ghost file system +

- used by Amazon web services, it's highly secure, elastic & efficient.
- works with Amazon's EC2, simple DB and S3

② Hadoop file system:- (HDFS)

- Block-structured system for distributed applications
- HDFS stores files in blocks across clusters.

③ Kosmos File System (KFS):-

- open source project, written in C++ by startup Kosmix.
- KFS distributed data across multiple servers, first splitting it into 64 MB block
- It is designed to quickly handle large file chunks across different servers.

7] Cloud Storage :-

→ cloud storage solves traditional storage issues by storing data online, managed by cloud service providers

→ Traditional storage problem solved by cloud

① Unpredictable data growth.

② High cost & complexity

③ Security.

→ cloud storage uses pay-as-you-go model.

→ cloud providers build flexible ~~information~~ infrastructure for all clients.

Key Attributes

→ Multi-tenancy :- multiple customers share the same physical server for data storage.

→ Resource pooling :- providers share large-scale resources among customers.

→ Scalable & elastic :- storage capacity adjusts based on data needs.

→ On-demand self-service :- Users can manage storage independently.

④ Provisioning Cloud Storage :-

- provisioning : setting up IT infrastructure and managing access to data & resources.
- cloud provisioning :- Allocating a cloud service provider's resources to a customer.
- Delivery models :- three main cloud provisioning models are:
 - i) Advanced
 - ii) Dynamic
 - iii) User self-provisioning.

cloud provisioning challenges :-

- ① complex management & monitoring
 - requires multiple tools to manage cloud resources
- ② Resource dependencies
 - cloud application often depend on computing, networking & storage resources
- ③ policy enforcement.
 - streamlining resource access while ensuring that unnecessary resources are not provided.

③ Cloud Storage Requirements:-

(i) multi-tenancy:-

- A cloud architecture where multiple users share resources in public or private clouds.
- multiple users can access the same application logic simultaneously.

(ii) Security:-

- Security policy defines a set of security rules & regulations.
- These policies further clarify how the rules are implemented & enforced.
- Data is secured using encryption, authentication & authorization.

(iii) Secure transmission channel:-

- methods to secure n/w communication include
 - ① virtual private n/w
 - ② HTTP
 - ③ Private n/w
 - ④ Transport layer security (TLS) and, secure socket layer (SSL).

(iv) Performance:-

- cloud storage performance depends on speed & latency.

2) cloud data management interface (CDMI) :-

- CDMI is the interface for managing data in cloud storage.
- It enables retrieving, creating, deleting & updating data from any cloud.
- It helps clients find available capabilities in cloud storage & manage data.

Features & Benefits :-

- ① Powerful management :- CDMI adds management features like user account administration, security access, monitoring & billing.
- ② Capability discovery :- Clients can discover cloud storage capabilities to manage containers & data.
- ③ Open standard :- CDMI is an ISO standard compatible with Openstack swift & Amazon S3 models.

Operations of CDMI :-

- ① Types :-
 - Ⓐ Those using CDMI content type in the HTTP body
 - Ⓑ Those not using the CDMI content type
- ② Management :- CDMI can retrieve & store data as well as manage it.
- ③ Control path :- Defines how data management is done, ensuring smooth data retrieval.
- ④ Capability management :- CDMI also manages cloud storage properties based on the capabilities available to the client.

① Data management for cloud storage.

- Data management includes acquiring, storing, protecting & processing data.
- It ensures data is validated & accessible to stakeholders when required.
- cloud data management is a part of modern data strategies, which focus on combining cloud & traditional data approaches.
- This unified approach uses the cloud's scalability and efficiency to provide faster insights.
- The SNIA cloud storage Technologies Initiative promotes standardized cloud storage adoption.
- It focuses on services, management & ensuring data portability across multiple clouds.

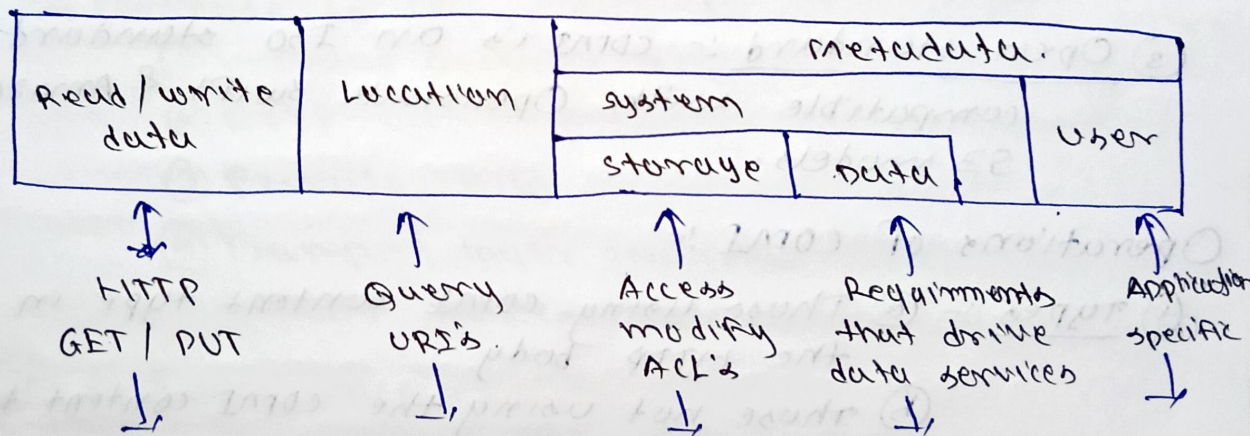


Fig:- SPDM model for cloud storage.

Usage:-

- It involves managing metadata for storage system, data & user.
- metadata types: storage system metadata, data system metadata, user metadata.
- contain info about access, control, encryption, data retention & more.